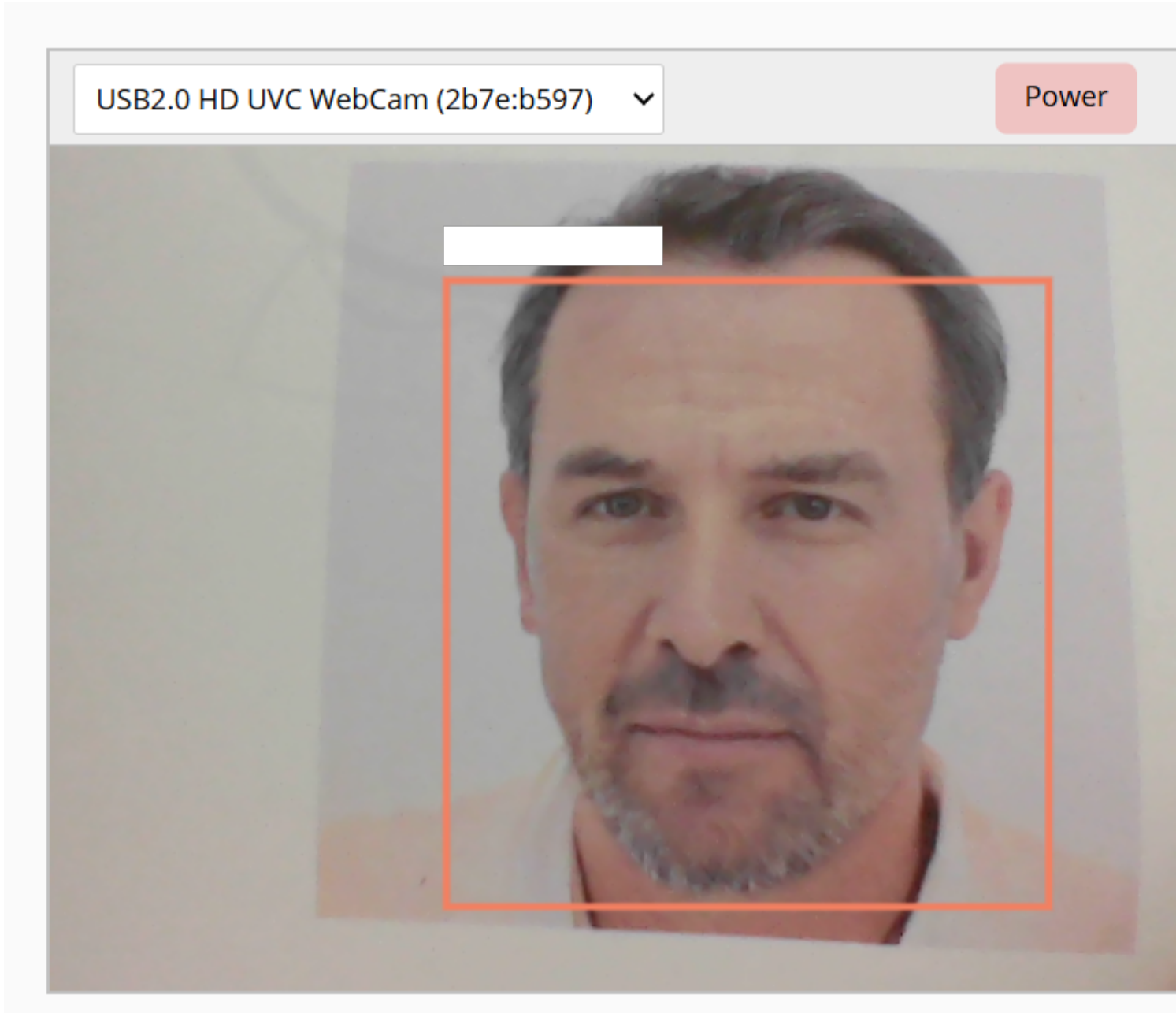


Open Dashboard

Open Dashboard is not a block that can be used in block coding, but it allows you to open a dashboard where you can check how the model used in the extension module is applied.

Dashboard Screen

When you click Open Dashboard, you can see the following screen.



Detailed Features

Power

Turns the selected camera on or off.

Load Model

Loads the trained age, gender, expression models. This is a required step to use the 'Face Expression' extension module.

Show Result

Displays the detected age, gender, and expression results on the camera screen.

Detect

Starts or stops age, gender, expression detection.

You can choose whether to run it once with the Once button or continuously with the Continuous button. You can also stop the detection using the Stop button.

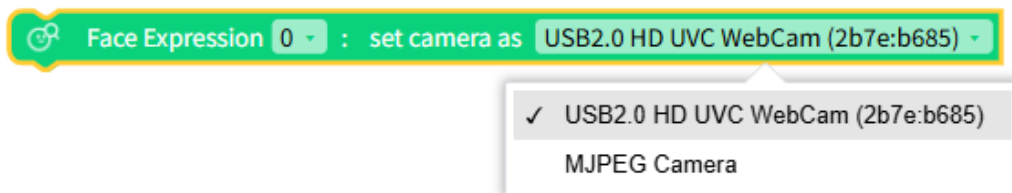
Sensory Data

Outputs the detected age, gender, expression detection data values.

Blocks

Set Camera

Select the camera to be used for the Face Expression module.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
camera	Dropdown option	Camera to use	List of available cameras

JavaScript Code

```
// Set a specific camera for face expression detection (id is an example)
$('FaceExpression*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb';
```

Python Code

```
# Set a specific camera for face expression detection (id is an example)
__('FaceExpression*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb'
```

Load Face Expression Model

Loads the trained age, gender, expression model.

This step is required to use the functions of the 'Face Expression' module.

Face Expression 0 : load face expression model | wait ✓

Load the trained face expression model. This task is necessary to use the functions of the 'Face Expression' module.

JavaScript Code

```
// Load age, gender, expression model | Wait 0
$('FaceExpression*0:load_model').d = 1;
await $('FaceExpression*0:!load_model').w();

// Load age, gender, expression model | Wait X
$('FaceExpression*0:load_model').d = 1;
```

Python Code

```
# Load face model | Wait 0
__('FaceExpression*0:load_model').d = 1
await __('FaceExpression*0:!load_model').w()

# Load face model | Wait X
__('FaceExpression*0:load_model').d = 1
```

Detect Age, Gender, Expression Once

Analyzes the face currently on the screen and displays the detected age, gender, and expression exactly once.



Detect the face on the screen and display age, gender, facial expression once.

JavaScript Code

```
// Detect age, gender, and expression once
$('FaceExpression*0:detect.once').d = 1;
```

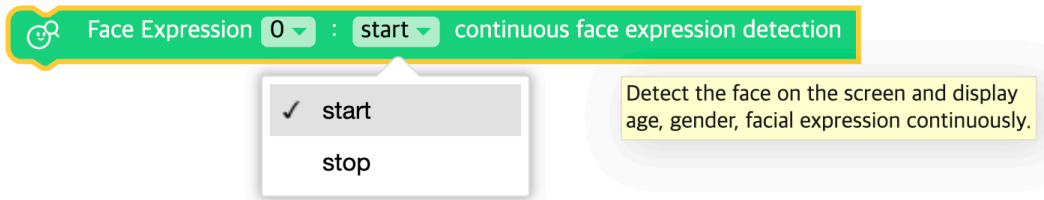
Python Code

```
# Detect age, gender, and expression once
__('FaceExpression*0:detect.once').d = 1
```

Detect Age, Gender, Expression Continuously

Starts or stops continuous age, gender, expression detection.

When continuous detection starts, it continuously follows the face currently on the screen and displays the detected results on the screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown option	Detect	Start (1), Stop (0)

JavaScript Code

```
// Start continuous age, gender, expression detection
$('FaceExpression*0:detect.continuous').d = 1;

// Stop continuous age, gender, expression detection
$('FaceExpression*0:detect.continuous').d = 0;
```

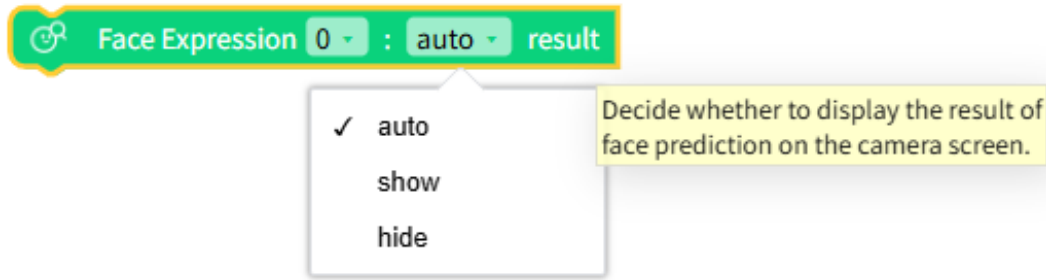
Python Code

```
# Start continuous age, gender, expression detection
__('FaceExpression*0:detect.continuous').d = 1

# Stop continuous age, gender, expression detection
__('FaceExpression*0:detect.continuous').d = 0
```

Show Age, Gender, Expression Detection Results

Determines whether to display the detected age, gender, and expression results on the camera screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown option	Detection results	Show (1), Hide (0)

JavaScript Code

```
// Show age, gender, expression detection results
$('FaceExpression*0:display').d = 1;

// Hide age, gender, expression detection results
$('FaceExpression*0:display').d = 0;
```

Python Code

```
# Show age, gender, expression detection results
__('FaceExpression*0:display').d = 1

# Hide age, gender, expression detection results
__('FaceExpression*0:display').d = 0
```

Age Detection Data

Returns the age detection result data.



JavaScript Code

```
// Age detection data  
$('FaceExpression*0:age').d;
```

Python Code

```
# Age detection data  
__('FaceExpression*0:age').d
```

Gender Detection Data

Returns the gender detection result data.

The returned data is 'Male', 'Female'.



JavaScript Code

```
// Gender detection data  
$('FaceExpression*0:gender.name').d;
```

Python Code

```
# Gender detection data
__( 'FaceExpression*0:gender.name' ).d
```

Expression Detection Data

Returns the expression detection result data.

The returned data is 'happy', 'sad', 'angry', 'fearful', 'disgusted', 'surprised', and 'neutral'.



expression

JavaScript Code

```
// Expression detection data
$( 'FaceExpression*0:expression.name' ).d;
```

Python Code

```
# Expression detection data
__( 'FaceExpression*0:expression.name' ).d
```

Is Gender Detected?

Returns **True (1)** / **False (0)** depending on whether gender detection has been made.



Whether the gender is detected

JavaScript Code

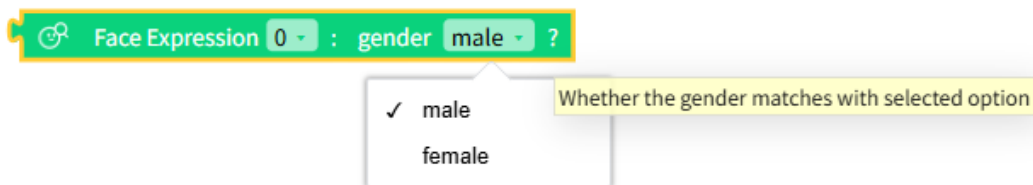
```
// Is gender detected?  
$('FaceExpression*0:gender.detected').d;
```

Python Code

```
# Is gender detected?  
__('FaceExpression*0:gender.detected').d
```

Is Gender ~ ?

Returns **True (1)** / **False (0)** depending on whether the selected gender matches the detected gender.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
gender	Dropdown option	Gender	male, female

JavaScript Code

```
// Is the detected gender male?  
$('FaceExpression*0:gender.name').d === 'male';  
  
// Is the detected gender female?  
$('FaceExpression*0:gender.name').d === 'female';
```

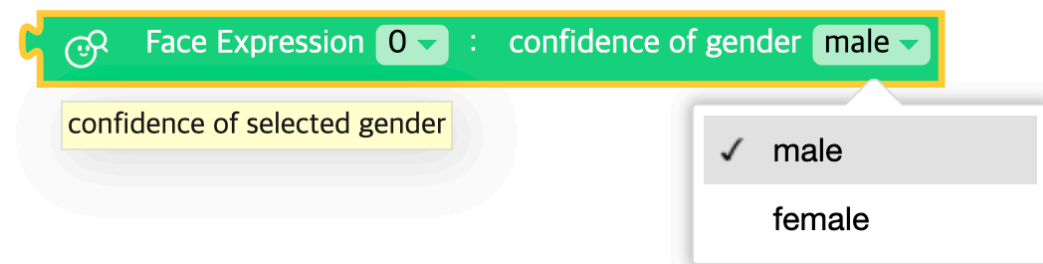
Python Code

```
# Is the detected gender male?
__('FaceExpression*0:gender.name').d == 'male'

# Is the detected gender female?
__('FaceExpression*0:gender.name').d == 'female'
```

Probability (Confidence) that Gender is ~

Returns the probability (confidence) that the selected gender matches the detected gender.
The return value is a real number between 0 and 1.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
gender	Dropdown option	Gender	Male(0), Female(1)

JavaScript Code

```
// Probability that the detected gender is male
$('FaceExpression*0:gender.confidence').d[0];

// Probability that the detected gender is female
$('FaceExpression*0:gender.confidence').d[1];
```

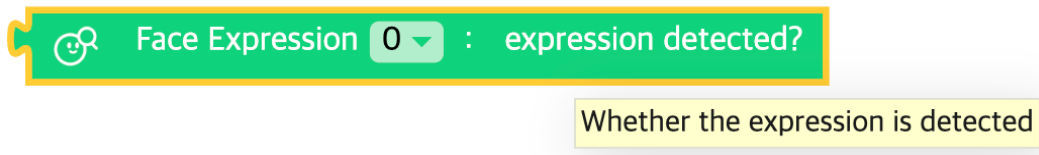
Python Code

```
# Probability that the detected gender is male
__( 'FaceExpression*0:gender.confidence' ).d[0]

# Probability that the detected gender is female
__( 'FaceExpression*0:gender.confidence' ).d[1]
```

Is Expression Detected?

Returns **True (1) / False (0)** depending on whether expression detection has been made.



JavaScript Code

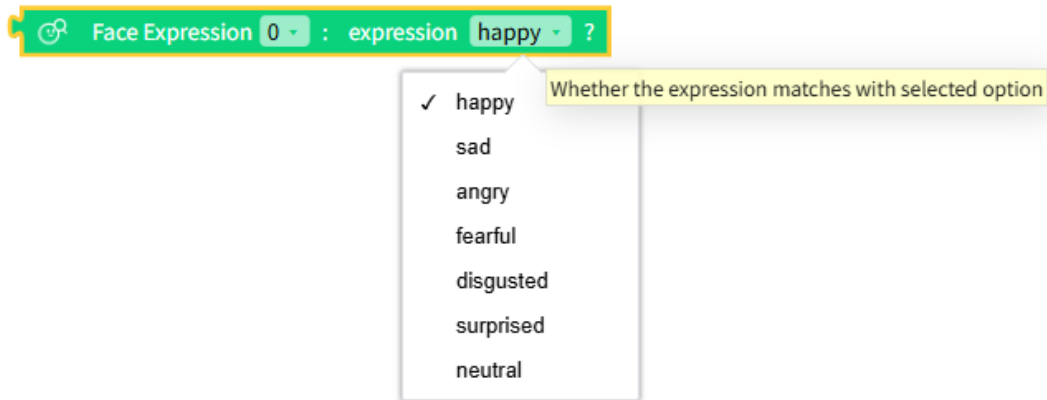
```
// Is expression detected?
$( 'FaceExpression*0:expression.detected' ).d;
```

Python Code

```
# Is expression detected?
__( 'FaceExpression*0:expression.detected' ).d
```

Is Expression ~ ?

Returns **True (1) / False (0)** depending on whether the selected expression matches the detected expression.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
expression	Dropdown option	Expression	happy, sad, angry, fearful, disgusted, surprised, neutral

JavaScript Code

```
// Is the detected expression happy?
$('FaceExpression*0:expression.name').d === 'happy';

// Is the detected expression sad?
$('FaceExpression*0:expression.name').d === 'sad';

// Is the detected expression angry?
$('FaceExpression*0:expression.name').d === 'angry';

// Is the detected expression fearful?
$('FaceExpression*0:expression.name').d === 'fearful';

// Is the detected expression disgusted?
$('FaceExpression*0:expression.name').d === 'disgusted';

// Is the detected expression surprised?
```

```
$('FaceExpression*0:expression.name').d === 'surprised';

// Is the detected expression neutral?
$('FaceExpression*0:expression.name').d === 'neutral';
```

Python Code

```
# Is the detected expression happy?
__('FaceExpression*0:expression.name').d == 'happy'

# Is the detected expression sad?
__('FaceExpression*0:expression.name').d == 'sad'

# Is the detected expression angry?
__('FaceExpression*0:expression.name').d == 'angry'

# Is the detected expression fearful?
__('FaceExpression*0:expression.name').d == 'fearful'

# Is the detected expression disgusted?
__('FaceExpression*0:expression.name').d == 'disgusted'

# Is the detected expression surprised?
__('FaceExpression*0:expression.name').d == 'surprised'

# Is the detected expression neutral?
__('FaceExpression*0:expression.name').d == 'neutral'
```

Probability (Confidence) that Expression is ~

Returns the probability (confidence) that the selected expression matches the detected expression. The return value is a real number between 0 and 1.

Face Expression 0 : confidence of expression happy

confidence of selected expression

- ✓ happy
- sad
- angry
- fearful
- disgusted
- surprised
- neutral

Dropdown Options and Input Values

Name	Type	Description	Range / Type
expression	Dropdown option	Expression	happy, sad, angry, fearful, disgusted, surprised, neutral

JavaScript Code

```
// Probability that the detected expression is happy
$('FaceExpression*0:expression.confidence').d[0];

// Probability that the detected expression is sad
$('FaceExpression*0:expression.confidence').d[1];

// Probability that the detected expression is angry
$('FaceExpression*0:expression.confidence').d[2];

// Probability that the detected expression is fearful
$('FaceExpression*0:expression.confidence').d[3];

// Probability that the detected expression is disgusted
```

```
$('FaceExpression*0:expression.confidence').d[4];

// Probability that the detected expression is surprised
$('FaceExpression*0:expression.confidence').d[5];

// Probability that the detected expression is neutral
$('FaceExpression*0:expression.confidence').d[6];
```

Python Code

```
# Probability that the detected expression is happy
__('FaceExpression*0:expression.confidence').d[0]

# Probability that the detected expression is sad
__('FaceExpression*0:expression.confidence').d[1]

# Probability that the detected expression is angry
__('FaceExpression*0:expression.confidence').d[2]

# Probability that the detected expression is fearful
__('FaceExpression*0:expression.confidence').d[3]

# Probability that the detected expression is disgusted
__('FaceExpression*0:expression.confidence').d[4]

# Probability that the detected expression is surprised
__('FaceExpression*0:expression.confidence').d[5]

# Probability that the detected expression is neutral
__('FaceExpression*0:expression.confidence').d[6]
```

Age, Gender, Expression Model Loading State Value

Returns the loading state of the age, gender, and expression model.

If it has not been loaded yet, it returns 0; if it is loading, it returns 1; if loading is complete, it returns 2.

 Face Expression : face expression model loading state

Returns the state of face expression model loading.
Returns 0 if it has not been loaded yet, 1 if it is on loading process, and 2 if the loading has completed.

JavaScript Code

```
// Age, gender, and expression model loading state value  
$('FaceExpression*0:model_state').d;
```

Python Code

```
# Age, gender, and expression model loading state value  
__('FaceExpression*0:model_state').d
```